Investigation report

Did newer signup cohorts convert to subscription at a lower rate?

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Data

PocketToons SQLite, synthetic reference fixture
April–August 2026 signup cohorts, UTC

Snapshot

2026-09-30 23:59:59 UTC (declared)
Build 3a0dbed21d2f45958aefa2887a0cadd8

Run

Independent investigation
2026-11-09
10 query attempts, including 1 error

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How to read this report

- Checks refer to numbered JSONL evidence logs stored alongside this report.
- Rates are cohort counts divided by eligible signups. Percentage-point changes differ from relative percentage changes.

Terms used in this report

Term	Meaning
Cohort	Users signing up in the same UTC calendar month.
Conversion	At least one qualifying successful subscription charge within 720 hours after signup; a later refund does not reverse conversion.

01. Chapter

The answer

01

Yes. Among fully observed signup cohorts, 30-day subscription conversion fell from 1.94% in May–June to 0.53% in July–August: a 1.40 percentage-point decline (72% relative).

- May–June: 94 of 4,854 signups; July–August: 29 of 5,426 (checks 4 and 9).
- The decline occurs within each major acquisition source, so a shift in source mix alone does not explain it (check 6).
- September has no eligible users and cannot be compared on this metric. The cause of the decline remains unproven (checks 4 and 6).

02. Chapter

How we investigated

02

What data we used

- Approved subscription_conversion_30d, funnel and lifecycle group v1.0. Window is [signup instant, signup instant + 30 days); a qualifying charge is positive, valid and successful. Eligible users must have the entire window within [2026-04-01, 2026-10-01) UTC.
- Gold cohort rows at month 0 supply the baseline; Silver profiles supply fixed acquisition-source slices. An independent Bronze recount using current registry SQL matches all five mature monthly counts (checks 4, 6, 9).
- Snapshot is the metadata assertion 2026-09-30 23:59:59Z; coverage is the catalog contract, not a verified freshness guarantee. The repository download has no Git metadata; files are dated 2026-11-09.

03. Chapter

What happened

03

The movement

Signup month	Eligible	Converted	30-day rate
April 2026	1,600	73	4.56%
May 2026	2,201	38	1.73%
June 2026	2,653	56	2.11%
July 2026	2,477	13	0.52%
August 2026	2,949	16	0.54%
September 2026	0	0	Not available

- April is the first observed coverage month, so May–June is the principal comparison baseline; the pattern is also evident versus April. The sequence is not strictly monotonic (check 4).
- September contains 1,720 signups but none with a complete 30-day window by coverage end; zero eligible does not mean zero conversions.

Who drove it

Source	May–June	July–August
Organic	46 / 2,389 = 1.93%	6 / 996 = 0.60%
Paid search	23 / 1,273 = 1.81%	5 / 907 = 0.55%
Paid social	23 / 1,144 = 2.01%	18 / 3,469 = 0.52%
Unknown	2 / 48 = 4.17%	0 / 54 = 0.00%
All	94 / 4,854 = 1.94%	29 / 5,426 = 0.53%

- Paid social share rose from 23.6% to 63.9% of eligible signups. Applying late source-specific rates to the early source mix gives 0.56%, close to the observed late 0.53%. This descriptive standardization supports a broad within-source decline; it is not a causal decomposition (check 6).
- Unknown attribution is small but retained; rates for small groups are unstable.

04. Chapter

Why it happened

04

Hypothesis	Status	Evidence
Newer cohorts converted less	Supported	Each July and August rate is lower than May and June (check 4).
Source mix alone explains decline	Rejected	All three major source rates fell; standardized late rate is 0.56% (check 6).
A particular product or campaign change caused the decline	Inconclusive	No causal experiment or change history was examined.

- Early stop after the baseline and one hypothesis round: the question is answered and a major composition explanation tested. Mechanisms and interventions remain open.

05. Chapter

Data quality and limits

05

Data quality

- Build succeeded 2026-11-09 11:16:56Z and recorded successful Gold conversion and Silver profile reconciliations. Twenty Bronze views, ten Silver tables and eight Gold tables exist (checks 1–3).
- Current registry SHA-256 4b5b2f0e881666570ace017c06ff1bca380fff83c8dfa679317706e6a8815554 differs from build-recorded 57b21d71027782fc28c8ea7eea4eb0dd9ed18273577d81bf00eaded632ff1227; builder hashes match. The direct source recount under the current definition matches Gold, but full source fingerprints were not freshly verified (checks 3 and 9).
- Profile and payment keys are unique and non-null; signup and success timestamps tested are valid; no first subscription precedes signup; no profile or payment rows are flagged excluded (check 8).
- Thirteen reviewed source amount corrections are flagged suspicious but usable; two users, one each in July and August, had a reviewed qualifying subscription charge in their window. Excluding at most both users gives 27/5,426 = 0.50%, leaving the conclusion unchanged; this is a conservative sensitivity bound, not a recalculation of alternate valid transactions (checks 7 and 10).
- Check 5 failed on an incorrect column name; the corrected check 8 succeeded. All successful results were untruncated.

Limitations

- This is a synthetic fixture and an observational comparison, not evidence of a real-world causal mechanism. The declared data coverage and build provenance do not independently guarantee source freshness.

– The question does not specify a window or baseline; the approved 30-day definition and adjacent mature May–June versus July–August cohorts are used. Other definitions or time windows may yield different answers.

06. Chapter

What to do next

06

01 Inspect signup-to-purchase journeys and product, pricing or campaign changes around July, preserving cohort and 30-day windows; test causal explanations before acting on them.

02 Refresh data through at least 30 days after September signups before comparing that cohort, and reconcile source fingerprints and registry provenance on a future validated build.

Appendix

Evidence appendix

A

Check	Purpose and file
1–3	Object inventory, dataset metadata and full build / reconciliation: 01_objects, 02_metadata, 03_build
4	Monthly cohorts: 04_cohorts
5, 8	Quality query error and corrected retry: 05_quality, 08_quality_retry
6–7	Acquisition slices and exceptions: 06_segment, 07_exceptions
9–10	Bronze recount and correction sensitivity: 09_source_recount, 10_review_impact